Audiorium: Cultivating Diversity in Music Recommendation with Latent Representations of Audio

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Introduction

- With the emergence of music streaming, recommendation algorithms has become a primary force in culture curation and global discovery [1, 6].
- Some audio features used in streaming service algorithms struggle to translate inclusively across genres and cultures.
- Demands for algorithmic optimization for engagement can create echo chambers of western pop dominance [2 , 7].

Research Question:

How can machine learning be utilized to develop a music recommendation algorithm that works beyond searching?

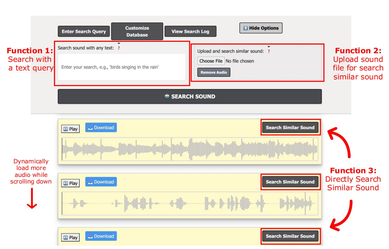
Dataset

Free Music Archive Small Dataset [3].

- The standard for genre classification and feature extraction.
- Features 161 genres, from broad (Rock) to niche (Glitch).
- Four Dataset Sizes: Small (8k tracks) - Full (106k tracks)
- Includes pre-computed audio features (MFCCs, Chroma) alongside artist and album data.
- Open License: Uses Creative Commons for raw audio research.

Methods

- Combine capabilities of:
  - CLAP (text-to-audio & audio-to-audio) [4]
  - ClapSearch (search interface) [5]
  - OpenClip (image-to-audio)
- Inspired by Spotify's Text2Track [8] and MusicMakerHub
- Rule-based recommendation adds slight randomization to top results.



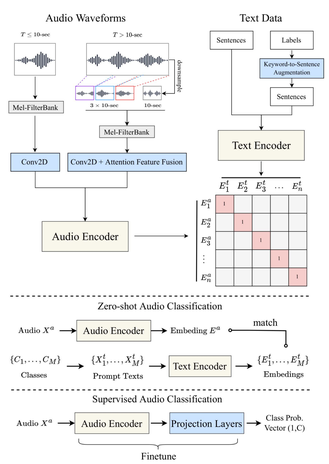


Figure 3. CLAP model (right) architecture and CLAPSearch interface (left) [4, 5]

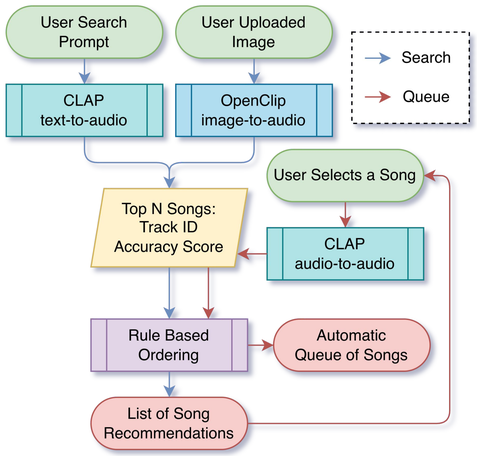


Figure 1. Search and queue retrieval flow for Audiorium.

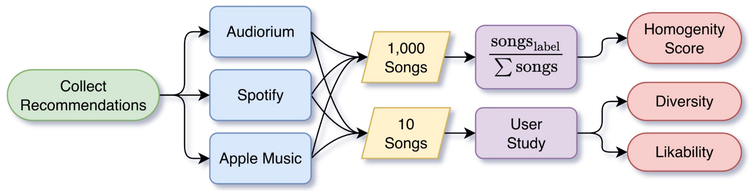


Figure 2. Objective and subjective evaluation method for Audiorium

Prototype

A simple UX prototype was developed to display results the algorithm. This interface was modeled to have aspects of LLM and music streaming interfaces in one app.

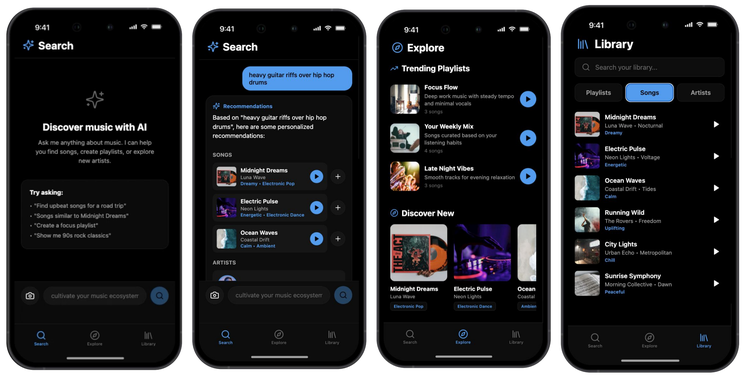


Figure 4. Demo interface created to show search and recomendation functions

Results

Objective

- Homogeneity score: percentage of recommendations that include songs tagged as major label artists.
- The ideal score would be 0.3, which is closely representative of the population of label vs non-label music in all datasets.

Audiorium	Spotify	Apple Music
0.28	0.61	0.83

Table 1: Homogeneity scores for 1,000 recommended songs in each algorithm

Subjective

- Participants listened to the recommendations generated by Audiorium's algorithm versus a other algorithms and rated diversity and likability on a Likert scale (1-5).

Measure	Audiorium	Spotify	Apple Music
Diversity	3.62	2.21	1.35
Likability	1.49	3.25	2.51

Table 2: Likert Scale means of user perception of diversity and likability of each algorithm

- Confounding variable: Audiorium's open source dataset vs other service's database of copyrighted and professionally produced music.

Future Work

- Retraining of Free Music Archive Large Dataset
- Longitudinal studies comparing commercial algorithms to Audiorium where users can use the platform for days to weeks.
- Create a portion of the experiment that quantifies the results from the image recommendation capability.

References

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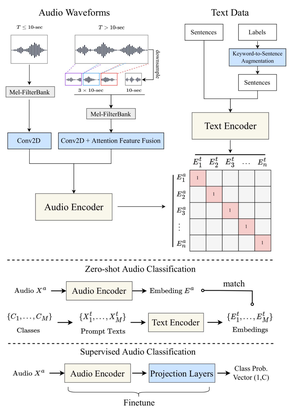

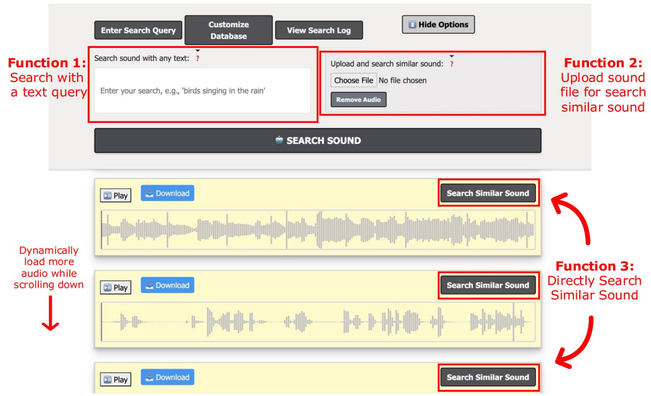

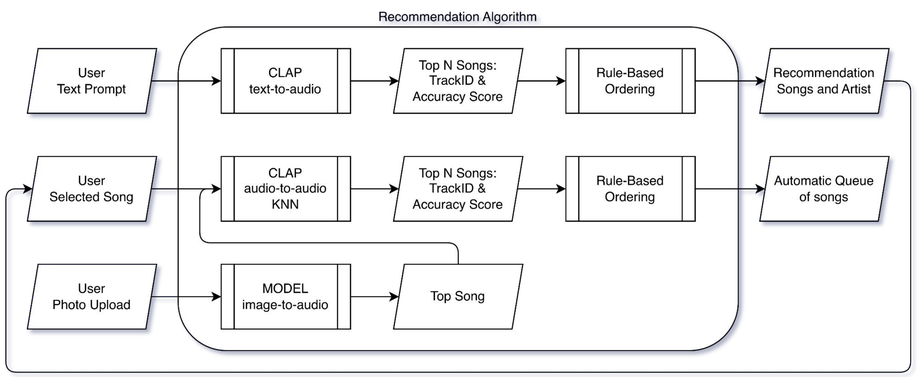

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